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Asset Management Planning

03

Operational Excellence, Glencore Copper
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Supported by the AM Network Pty Ltd and Work Management Solutions Pty Ltd

Asset Management Planning

Preface

The Asset Management Framework, Policy and Manuals are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source the commodities that advance everyday life. This document lays out the minimum standard for Glencore Copper and compliance to this manual will demonstrate a maturity value equal to Planned/Deployed (3) on the Glencore Copper Asset Management Maturity Scale.

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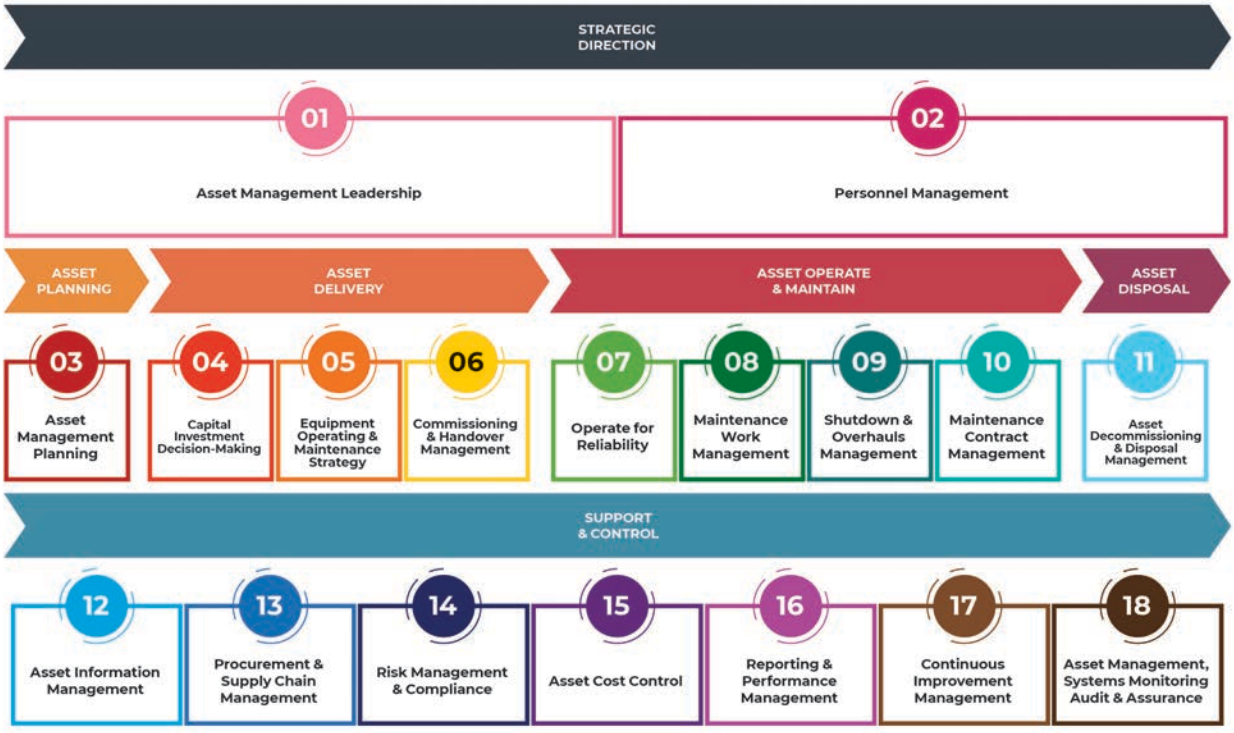
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1. Introduction

The Asset Management Policy and Manuals define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring assets are capable of meeting business and stakeholder objectives.

Figure 1-1 describes the high-level Asset Management Framework, which enables Glencore Copper to deliver its Asset Management purpose and vision.

FIGURE 1-1 ASSET MANAGEMENT FRAMEWORK



2. Objectives

The objective of this manual for Asset Management Planning is to define how Asset Management Planning will be delivered across Glencore Copper and it also considers the content of an Asset Management Plan as well as the implementation, improvement and monitoring of the plan.

Asset Management Plans describe in detail the activities to be carried out on assets and how to coordinate them in order to deliver the Asset Management Strategy and Objectives in line with the Asset Management Framework.

The Asset Management Plan should be developed at a level appropriate for each region and/or site. The minimum level expected would be for plans to be developed for each major processing area (for example crushing, grinding, concentrating, roasting, smelting, or leaching) or mobile fleet type (for example hauling fleet). The focus should be on processing areas and/or mobile fleets that are critical to production or constrain production, pose high or material risk to the business and are cost intensive to operate and maintain.

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2.1 Accountabilities

Dedicated resources must be assigned and capabilities developed within each Operation / Process Area to deliver Asset Management Planning processes that enables effective delivery of the Asset Management objectives.

- Assign accountability for each or all the following:
  - Asset Management Plan development
  - Element E05 Equipment Operating and Maintenance Strategy
  - Asset Management Planning strategies defined and documented
  - Implementing the Asset Management Plan
  - Compliance to the Asset Management Plan (review, audit and monitor).
  - Improvement of the Asset Management Plan (update)
- Define and document Asset Management Planning competencies required for development, implementation, improvement and monitoring of the Asset Management Plans.
- Plan, deliver and record Asset Management Planning training to support the embedment of Asset Management Plan competence and knowledge.

2.2 Scope

This manual is to be read in conjunction with the following manuals and documents:

- E01 Asset Management Leadership
- E02 Personnel Management
- E04 Capital Investment Decision-Making
- E05 Equipment Operating and Maintenance Strategy
- E07 Operate for Reliability
- E09 Shutdowns and Overhauls Management
- E11 Asset Decommissioning and Disposal
- E14 Risk Management and Compliance
- E16 Reporting and Performance Management
- E17 Continuous Improvement Management
- E18 Asset Management Monitoring, Audit and Assurance
- Strategic Asset Management Plan (SAMP)
- SAMP Development Guideline
- Asset Budget Attainability Guideline
- Life of Asset Plan (LoA)
- Asset Criticality Analysis Standard

The scope of Asset Management Planning covers the following:

- Principles, Drivers and Uncertainties
  - Defines the overview, objectives and approach for dealing with uncertainties that could dictate the decisions Glencore Copper makes regarding the management, construction and operation of assets.
- Organisational Arrangement for Asset Grouping
  - How Glencore Copper sites are structured and the roles and responsibilities to deliver the Plan and the activities of the Plan.
- Asset Coverage
  - What assets within the process area are covered by the Plan and the “as is” position in respect of the number, distribution, function, performance, condition, criticality and the age profile of the asset group.
- Objective and Performance Measure of the Asset Group
  - Explains the objectives, performance measures and trends.
- Asset Information

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  - Explains the framework for monitoring continuous improvement in asset management performance, through the collection, analysis and reporting on the data that is necessary to manage asset condition and performance.
- Gap Definition
  - Explains the gap between the "as is" and where the performance of the asset needs to be to deliver the business objectives.
- Planning Activities
  - How the gap between "as is" and "to be" is to be achieved.
- Justification
  - Provides clear auditable justification for all planned activities.

3. Definitions

Asset – a thing or entity that has potential or actual value to an organisation.

Asset Management Capability - ability to identify and embed a competency framework relevant to the Asset Management functions.

Asset Management Objectives – results to be achieved with respect to asset management.

Asset Management Plan (AMP) - provides documented information that specifies the following, that are required for an asset class (or individual asset or asset system), to achieve the organisation's asset management objectives:

- Activities - such as maintenance tasks, operational tasks and capital projects
- Resources - such as service contractors, information and money
- Timescales - along a planning horizon

Capital Expenditure – expenditure used to create new assets, renew assets, expand or upgrade assets or to increase the capacity of the existing assets beyond the original design capacity. CAPEX increases the value of the asset stock.

Creation/Acquisition/Upgrade Plan – works that are required to increase the capacity of the systems/asset to meet demand and levels of service.

Disposal Plan - includes any activity associated with disposal of a decommissioned asset including sale, demolition or relocation.

Line of Sight – in the context of asset management, line of sight refers to

- an approach within an organisation that looks to line up the work that is done directly on assets with the objectives of that organisation
- a discipline which recognises, accommodates and aligns the risk of owning a particular asset with the goals of the organisation that operates the asset

Renewals and Replacement Plan – Renewal and replacement expenditure is major work which does not increase the asset's design capacity but restores, rehabilitates, replaces or renews an existing asset to its original or lesser required service potential.

Strategic Asset Management Plan (SAMP) – documents and specifies how the organisational objectives are to be converted into Asset Management objectives. The approach for developing the Asset Management Plans and the role of the Asset Management System in supporting the achievements of the Asset Management objectives. The SAMP contains key information from the Asset Management Plans (AMPs), summarising the asset physicals, expenditures and initiatives that contribute to the organisational objectives.

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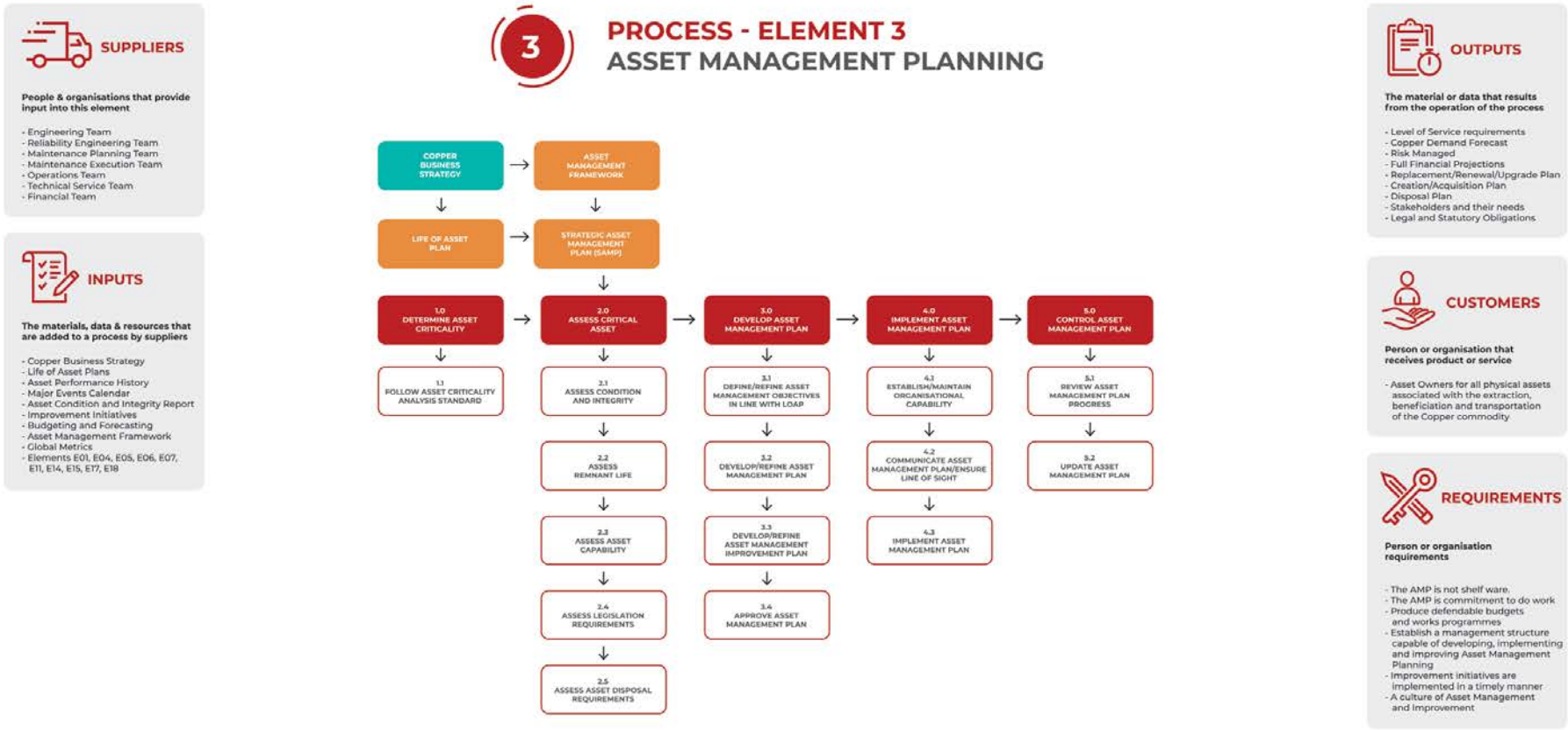
4. Approach

The approach for Asset Management Planning is described through:

- Process Flow - pictorially illustrates the key Asset Management Planning activities, refer to Figures 4-1.
- Process Description – defines in more detail the key Asset Management Planning activities, refer to Section 4.2.
- Process RACI Matrix - indicates for each activity, the positions who are Responsible, Accountable, who are Consulted prior to final decision/action, and those who are Informed of any outcomes and actions, refer to Section 4.3.

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FIGURE 4-1 ASSET MANAGEMENT PLANNING PROCES



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TABLE 4-1 ASSET MANAGEMENT PLANNING PROCESS DESCRIPTION

<table>
<tr><th>PROCESS STEP</th><th>PROCESS DESCRIPTION</th><th>DESCRIPTION</th></tr>
<tr><td>1.0</td><td>Determine Asset Criticality</td><td>Asset Criticality to be analysed and Criticality Ranking A, B or C assigned.</td></tr>
<tr><td>1.1</td><td>Follow Asset Criticality Analysis Standard</td><td>Follow the Glencore Copper Asset Criticality Analysis Standard to determine Asset Criticality. The focus should be on processing areas and/or mobile fleets that are critical to production or constrain production, pose high or material risk to the business and are very costly to operate and maintain.</td></tr>
<tr><td>2.0</td><td>Assess Critical Assets</td><td>Performing assessments on existing critical assets Ranked A to confirm they are fit-for-purpose and able to achieve asset management goals.</td></tr>
<tr><td>2.1</td><td>Condition and Integrity Assessed</td><td>Asset condition and integrity assessments to be conducted on critical assets (A). Refer to the critical asset register or Asset Management Plan/s for the critical assets to be assessed.

Condition and integrity assessment cover both the performance and the condition of the asset.
• Integrity – how the asset looks aesthetically and the assessed ability to function as desired. Typically associated to static assets such as tanks, pressure vessels and the like.
• Condition – understanding of degradation mechanism, how the asset is performing functionally and whether the level of performance is within desired parameters.

Generally, Asset condition is measured using a scaler approach such as the example below:
<table>
<tr><th>RATING</th><th>STATUS</th><th>DEFINITION OF RATING/CONDITION</th></tr>
<tr><td>1</td><td>Excellent</td><td>• No defect or only a slight decline
• As new condition and appearance</td></tr>
<tr><td>2</td><td>Good</td><td>• Minor defects or some early stages of deterioration evident but minor and does not cause serviceability problem
• Superficial wear and tear
• Some deterioration to finishes
• Major maintenance not required</td></tr>
<tr><td>3</td><td>Fair</td><td>• Average condition
• Moderate defects
• Moderate wear and tear that require maintenance
• Services are functional but need attention
• Deferred maintenance works exists</td></tr>
<tr><td>4</td><td>Poor</td><td>• Significant wear and tear
• Potential structural problems
• Major defects
• Components fail frequently</td></tr>
<tr><td>5</td><td>Unserviceable</td><td>• Asset has failed
• Not operational
• Not viable
• Unfit for normal use</td></tr>
</table></td></tr>
<tr><td>2.2</td><td>Assess Remnant Life</td><td>Determine Life Cycle Cost and Remaining Life Analysis to determine economical end of life and replacement plans. This can be achieved in the following steps:</td></tr>
</table>

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- Determine end of life – using a combination of design life, installation date, asset condition index and estimated remaining life. The following table provides an example of how this may be achieved.

Asset	Description	Install Date	Design Life (Years)	Current Date	Design Remaining Life (Days)	Condition (1-5)	Actual Remaining Life (Days)	Renewal Date
PMP001	Pump 01	01.01.2010	20	24.04.2021	3285	4	1825	22/04/2026

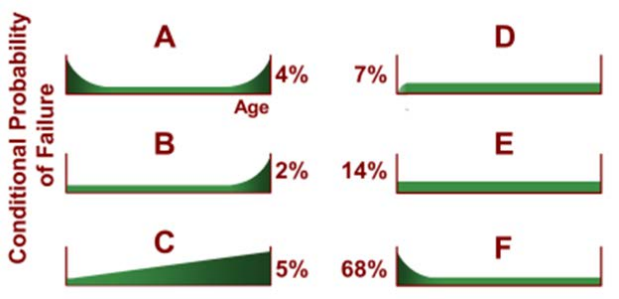
Remaining life can be estimated based on the below condition and remaining life criteria. Remaining life is estimated based on condition:

- Condition 1 = 100%
- Condition 2 = 75%
- Condition 3 = 50%
- Condition 4 = 25%
- Condition 5 = 0%

For example, a pump that was installed in 2010 with a design life of 20 years is condition assessed in April 2021. The expected design life remaining is 9 years, however its condition was assessed as a 4 which equates to 25% of design life remaining from above, therefore the estimated replacement/renewal date in October 2023, as opposed to January 2030. The intent of this process is to extend the replacement/renewal date beyond design life (subject to risk assessment) by ensuring good asset management practises are in place.

The model assumes linear degradation of asset condition, other degradation models should be considered as each site's asset management maturity improves and data becomes available to support such analysis. For example, the six (6) failure patterns in Reliability Centred Maintenance could be considered subject to available data and high confidence in that data:

A. Bathtub Curve – assets show three failure zones - infant mortality, random and wear out.
B. Wear Out – the asset shows clear wear out failures which can be highly predictable.
C. Fatigue – asset show increase in probability of failure over time.
D. Initial Break in – asset show no initial failure then constant probability of failure over life.
E. Random – asset show constant probability of failure.
F. Infant Mortality – asset shows early life failures.



Remaining life estimates can then be used as part of the Life Cycle Analysis process. Once remaining life is established this will provide an input into the budgeting process to ensure funds are available as analysed.

2.3	Assess Asset Capability	Assets must be capable of meeting production demand. Production demand is assessed and determined as part of Element E01 Asset Management Leadership and derived from the Business Strategy and Life of Asset Plan. As production demand changes, asset capability must be assessed and modified as required. This could involve: • Increase or decrease in asset base.

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<table>
<tr><td></td><td></td><td>

- Removing constraints
- De-bottlenecking
- Replacing asset with newer technology and greater efficiency
- Improving reliability, availability and utilisation performance.

</td></tr>
<tr><td>2.4</td><td>Assess Legislation Requirements</td><td>All relevant laws, regulations, standards and codes regarding the management of assets shall be adhered to. As these requirements change then actions need to be put in place to address these changes and managed through the SAMP and AMP development/improvement process.</td></tr>
<tr><td>2.5</td><td>Assess Asset Disposal Requirements</td><td>Refer to Element E11 Asset Decommissioning and Disposal Management

Assets are to be identified for disposal and a summary provided, including information regarding:
• What asset
• Reason for disposal
• When to dispose
• Disposal expenditure and revenue (salvage value) if to be sold.
• Inventory Reconciliation
• Enterprise Resource Planning (ERP) system reconciled, i.e. SAP</td></tr>
<tr><td>3.0</td><td>Develop Asset Management Plan</td><td>Developing and accessing asset management plans and initiatives to confirm their effectiveness to achieve the business plan.</td></tr>
<tr><td>3.1</td><td>Define/Refine Asset Management Objectives in Line with LoAP</td><td>Asset Management objectives shall be defined/refined in line with the Life of Asset Plan and the Strategic Asset Management plan. Asset Management objectives are then to be translated into asset management activities and detailed within an asset management plan/s.</td></tr>
<tr><td>3.2</td><td>Develop/Refine Asset Management Plan/s</td><td>To align with Glencore Copper values, Asset Management Plans should not be complex in nature and deliver pragmatic solutions. With that in mind Asset Management Plans for fixed plant shall be developed at Level 4 or 5 of the Glencore Copper Functional or Level 1 of Equipment Location Structure. This choice is dependent on process/system complexity and material risk at the level in the hierarchy. For mobile equipment, plans shall be developed at level 5 of the Functional Location Structure.

Refer to Master Data Design Standards - Functional Location and Equipment – located on the AMF hub.

<table>
<tr><td>FLOC/Equipment Level</td><td>Fixed Plant Example</td><td>Mobile Fleet Example</td></tr>
<tr><td>FLOC Level 1 – Site</td><td>EHM</td><td>EHM</td></tr>
<tr><td>FLOC Level 2 – Core Process</td><td>Processing</td><td>Mining</td></tr>
<tr><td>FLOC Level 3 – Main Process</td><td>Concentrating / Smelting / Refining</td><td>Open Cut / Underground</td></tr>
<tr><td>FLOC Level 4 – Sub-Process</td><td>Grinding</td><td>Loading</td></tr>
<tr><td>FLOC Level 5 – Sub-Process</td><td>Primary Grinding</td><td>Shovel Fleet / Loader Fleet</td></tr>
<tr><td>Equipment Level 1 – Maintainable Unit</td><td>SAG Mill</td><td>SH01 / LD201</td></tr>
<tr><td>Equipment Level 2 – Maintainable Function</td><td>Lube System</td><td>Boom / Engine</td></tr>
</table>
</td></tr>
</table>

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Equipment Level 3 – Maintainable Function	Pump	Boom Point Sheave / Fuel System

The Asset Management Plan/s should be sufficiently detailed to be appropriate for Glencore Copper and presented in a manner that engages the relevant stakeholders.

The below provides the minimum requirement for inclusion in the Asset Management Plan/s:

Heading	Sub-Heading and Contents
Scope	• Purpose of the plan • Description of the assets that the plan covers, what they do, how many and where they are. • Duration of the plan
Principles and Objectives	• References back to policies and strategy (LoA, SAMP etc.) • Objectives • Assumptions
Current Situation	• Risks identified through Risk Management process such as: o Risk management plans o Bow-ties o Business risk registers – major hazard and local o Business continuity plans o Critical controls • Historic and current performance levels • Asset Criticality (refer to Glencore Copper Asset Criticality Standard) • Replacement Value of Assets • Maintenance Plan and Cost • Operating Costs • Gaps against current targets and standards o Previous plan performance. o Are assets meeting demand, reliable and safe? o Compliance to legal and statutory requirements. o Profitability • Sustaining Capital (CAPEX) o Tabulate the list of current projects o Justification for investment
Future Demand	• LoA requirements and Strategic Plan • Future performance levels could include: o Throughput o Recovery o Reliability o Availability o Quality o Obsolescence o Technology Changes o Laws, Statutes and Standards • Current gap against future targets and standards • Future CAPEX needs (list) and justification
Solutions and Options	• Short, Medium and Long-Term options o Capital requirements. o Total Cost of Ownership for options o Risk and priority of options o Alignment to LoA and SAMP • Options analysis to meet demand. • Basis by which options are prioritised

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		Agreed Plan	The agreed plan provides summaries regarding the following items: • Operations and maintenance strategy • Major Maintenance and Shutdown Plan • Routine Maintenance Plan • Renewal/Replacement Plan • Acquisition/Upgrade Plan • Disposal Plan • Labour Forecast • Special Training and Professional Development Requirements • Total Costs and Benefits of Agreed Plan • Resources, constraints and risks to deliver the plan • Specialist training and professional development needs
		Roles and Responsibilities	• Owner and Approver of the Plan • Responsibility for developing, reviewing and updating of the plan, including frequency • Responsibility for delivery of the plan • Who are the stakeholders and how is the plan to be communicated?
		Performance Management	• Measures to monitor delivery against the plan • Measures to assure expected benefits are realised
		Reference and Supporting Documents	• LoA • SAMP • Capital Management Framework • Asset Management Framework • Risk Management Framework
3.3	Develop/Refine Asset Management Improvement Plan	Asset Management Improvement Plans are to be detailed, agreed and resourced to close the gaps in organisational asset management capability as identified in the Asset Management Maturity Self-Assessment (AMMSA).	
3.4	Approve Asset Management Plan	The Asset Management Plan is to be approved by the accountable person.	
4.0	**Implement Asset Management Plan**	**Establishing the capability to manage the workload to deliver the asset management plans.** Refer to Element E02 Personnel Management.	
4.1	Establish/Maintain Organisational Capability	Skills and experience to deliver the Asset Management Plan/s are to be identified and resources are to be engaged. Key considerations when determining the structure and capacity of the Asset Management resources include the need to focus on both current and future direction and the internal and external specialists and support staff needed to deliver the Asset Management objectives. While there may not be specific Asset Management roles designated at some Glencore Copper sites the responsibility for Asset Management delivery can sit with other functional resources such as the Maintenance Manager. The key message is that Asset Management is an organisational responsibility and as such everyone has a part to play in its success. Termed "Line of Sight" meaning that everyone within the organisation is aware of the organisational strategic direction and what role they play in the delivery and improvement of that strategy. As determined by E02 Personnel Management Section 2.0 and 3.0, resources are to be selected and personal development programmes are to be put in place to meet current and future business needs.	
4.2	Communicate Asset Management Plan/Ensure Line of Sight	The Asset Management Plan/s shall be understood by everyone that has a role to play in delivering the activates derived in the plan/s. The intent of the information contained within is to be widely communicated within each site and line of sight established.	

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4.3	Implement Asset Management Plan	Identified resources are to implement the Asset Management Plan/s to enable: • Delivery of identified asset management activities • Inform on risks and risk mitigation • Inform on change in production demand • Instigates a change in maintenance strategy and tactics • Instigates a change in shutdown strategy • Informs on Operational and Capital expenditure commitment
5.0	**Control Asset Management Plan**	**When the Asset Management Plan is changed it shall be controlled through a formal change management process and resources are to be identified and made responsible for the change management process.**
5.1	Review Asset Management Plan Progress	It is important to monitor the progress of the Asset Management Plan to ensure the asset management activities are delivering on the Asset Management objectives. The progress review should consider: • Planned v Actual Costs • Progress to work plan • Plan v Actual Performance Levels • Plan v Actual Risk Performance • Resource effectiveness – is everyone playing their part? • Impediments to the Plan and Actions to Mitigate • Improvement Opportunities Changes to the plan shall be communicated to key stakeholders, who in turn should communicate back to their teams.
5.2	Update Asset Management Plan	Asset Management Plans to be reviewed and updated annually in line with LoA, SAMP and budgeting reviews, or when significant business events occur such as: • Change in demand • Change in LoA • Change to the SAMP • Change in Risk • Change in Asset Performance (availability, MTBF, utilisation) • Change in costs • Technology advances • Change in law, standards, regulations An external review by an independent Asset Management Subject Matter Expert should also be considered.

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4.1 Process RACI Description

The RACI matrix below in Table 4-7 addresses the high-level functional elements of the Asset Management Planning process. For the overall Asset Management RACI matrix, refer to the Asset Management Intranet page.

Note: The HSE and Commercial Departments do not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE 4-2 PROCESS RACI

R	Responsible	The "doer" – the individual(s) who will perform an activity or directs others to perform the activity.
A	Accountable	The "owner" –the individual who ultimately "owns" the activity – there should only be one person
C	Consulted	The individual(s) that needs to provide feedback and contribute to the activity
I	Inform	The individual(s) who needs to be informed after a decision or action is taken if relevant. (via 1-way communication)

Function / Process Step	General Management	Maintenance/ Asset Management	Maintenance Execution	Maintenance Planning	Technical Services	Operations	Engineering	Finance	Supply
1.0 Determine Asset Criticality									
1.1 Follow Asset Criticality Analysis Standard		A	C	C		C	R		
2.0 Assess Critical Assets									
2.1 Assess Condition and Integrity		A	C	C			R		
2.2 Assess Remnant Life		A	C	C			R		
2.3 Assess Asset Capability		A					R		
2.4 Assess Legislation Requirements	I	A	C	C	C	C	R		
2.5 Assess Asset Disposal Requirements	I	A	C	C	C	C	R	C	C
3.0 Develop Asset Management Plan/s									
3.1 Define/Refine Asset Management Objectives in Line with LoAP	C	A	C	C	C	C	R	C	C
3.2 Develop/Refine Asset Management Plan	I	A	C	C	C	C	R	C	C
3.3 Develop/Refine Asset Management Improvement Plan	I	A	C	C	C	C	R	C	C
3.4 Approve Asset Management Plan	A	R							
4.0 Implement Asset Management Plan									
4.1 Establish/Maintain Organisational Capability	A	R	R	C	R	R	R	C	R
4.2 Communicate Asset Management Plan/Ensure Line of Sight	C	A	C	I	I	I	R	I	I

Function / Process Step	General Management	Maintenance/ Asset Management	Maintenance Execution	Maintenance Planning	Technical Services	Operations	Engineering	Finance	Supply
4.3 Implement Asset Management Plan	A	R	R	R	R	R	R	C	R
5.0 Control Asset Management Plan									
5.1 Review Asset Management Plan Progress	I	A	C	C	C	C	R	C	C
5.2 Update Asset Management Plan	I	A	C	C	C	C	R	C	C

TABLE 4-3 FUNCTIONAL DESCRIPTIONS

Function	Description
General Management	• Ensures Asset Management Plans are aligned to the business objectives. • Ensures asset management plans are approved and implemented in line with business objectives. • Ensures there are capable, competent and qualified personnel to develop and implement asset management planning. • Provides input with regarding the communication of asset management planning and asset management plans.
Maintenance Management	• Ensures asset criticality is assessed and implemented. • Ensures critical assets are assessed for: ○ Condition ○ Remaining life ○ Capability ○ Regulatory/Legal compliance ○ End of life and disposal requirements • Ensures asset management and asset management improvement plans are developed / refined. • Responsible for approval and implementation of asset management plans. • Responsible for ensuring capable, competent and qualified personnel develop and implement asset management plans. • Ensures asset management plans are reviewed and progress is in line with the plan. • Ensures asset management plans are updated.
Maintenance Planning	• Provides input into asset criticality analysis, asset condition and integrity, legal compliance and asset disposal requirements. • Provides input regarding capabilities and competencies for asset management planning and development and implementing asset management plans. • Responsible for the implementation of asset management plans. • Provides input into the review and update of asset management plans.

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Maintenance Execution	• Provides input into asset criticality analysis, asset condition, remaining life, legal compliance and asset disposal requirements. • Execution of plans in timely manner (if plans are not implemented timely, then failure cannot be prevented-just an opinion) • Provides input into the development / refinement of asset management and asset management improvement plans. • Responsible for ensuring capable, competent and qualified personnel to develop and implement asset management plans. • Responsible for the implementation of asset management plans. • Provides input into the review and update of asset management plans.
Operations	• Provides input regarding legal and asset disposal requirements. • Provides input into the development / refinement of asset management and asset management improvement plans. • Responsible for the implementation of asset management plans. • Responsible for ensuring capable, competent and qualified personnel develop and implement asset management plans. • Provides input for the review and update of asset management plans
Engineering	• Responsible for Asset Criticality analysis, asset condition, remaining life, legal compliance and asset disposal requirements. • Responsible for the development / refinement of asset management and asset management improvement plans. • Responsible for ensuring capable, competent and qualified personnel to develop and implement asset management plans. • Responsible for the implementation of asset management plans. • Responsible for the review and update of asset management plans.
Finance	• Provides input into asset disposal requirements. • Provides input into the development / refinement of asset management and asset management improvement plans. • Provides input into asset management plan implementation. • Provides input into asset management plan review and update.
Technical Services	• Provides input into asset disposal and legal compliance requirements. • Provides input into the development / refinement of asset management and asset management improvement plans. • Responsible for ensuring capable, competent and qualified personnel develop and implement asset management plans. • Responsible for the implementation of asset management plans. • Provides input into Asset Management plan review and update

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Supply	• Provides input into asset disposal and legal compliance requirements. • Provides input into the development / refinement of asset management and asset management improvement plans. • Responsible for ensuring capable, competent and qualified personnel develop and implement asset management plans. • Responsible for the implementation of asset management plans. • Provides input into Asset Management plan review and update • Ensure delivery and quality of part to maintain optimum inventory

5. Deployment

The deployment of the Asset Management Planning approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching and mentoring

5.1 Awareness Presentation

This Manual is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to the Asset Management Framework and business success.

5.2 Position Description Alignment

To effectively deploy this Manual, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Copper.

The RACI matrix helps ensure everyone is working towards the same goal. It clarifies, not only the individual's responsibilities, but also that of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including KPIs need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

5.3 Competency Assessment

This Manual is supported by an Asset Management competencies assessment tool which is used to:

- Guide recruitment, selection, development and deployment of people in asset management roles across the company
- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance.
- Map, build and manage the asset management talent pool across the company.
- Identify strengths and development needs in asset management practitioners.

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5.4 Training, Coaching and Mentoring

Coaching and mentoring are another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving performance in the short term where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring is required when a person is not currently working competently in their role. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

6. Measures of Success

Refer to Glencore Copper's AM Manual E16 Reporting and Performance Management Manual for current KPI's

TABLE 6-1 MEASURE OF SUCCESS

Metric	**Critical Assets Registered in CMMS (SAP)**
Definition	All critical assets (A) are identified and registered in SAP and criticality score assigned.
Performance Indicator	100% critical asset in SAP with criticality score.
Calculation	
Rationale	Identification and management of critical assets provides assurance that Glencore Copper can deliver on the asset management objectives. These critical assets having been identified as such play a significant part in achieving the objective and are to be identified, allocated a criticality ranking and registered in SAP as criticality type A assets.
Action	If significant business impact occur that are within the threshold of critical asset classification in the Glencore Copper Asset Criticality Analysis manual, then all asset responsible for that impact should be checked to see if criticality has been assigned in SAP and that it is correct.

Metric	**% Critical Systems with Asset Management Plan**
Definition	Critical systems identified in SAP are to have an asset management plan (AMP) developed for that system or be incorporated into an asset management plan of a parent system.
Performance Indicator	100% critical systems with AMP.
Calculation	Number of critical systems with AMP/Number of critical systems.
Rationale	In order to be able to manage critical systems to ensure asset management objectives are achieved, asset management plans are to be developed for these systems so that asset management activities are developed, managed and improved.
Action	Develop AMPs for critical systems.

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Metric	Critical Asset Condition
Definition	Critical asset condition and integrity register in place and updated within the last 12 months.
Performance Indicator	100% of critical assets have condition assessed and registered.
Calculation	Number of critical assets with condition assessment/Number of critical assets.
Rationale	Understanding the condition of critical assets provide a means by which remaining life can be estimated and the investment and effort required to manage those assets and to enable forward planning to occur for asset replacement/refurbishment.
Action	Conduct asset condition assessment and record in register. Develop action items as required.

Metric	Top 5 Losses
Definition	Identifying and managing top 5 business losses.
Performance Indicator	Top 5 losses identified.
Calculation	Pareto analysis across multi criteria operational losses.
Rationale	Understanding the top business losses provides data that will enable effort to be focused on the greatest impacts on the business. Provides focus for root cause analysis effort to ensure that asset utilisation targets are met or exceeded.
Action	Conduct root cause failure analysis for top 5 business impacts. Implement action plans to eliminate or significantly reduce loss to an acceptable level.

Metric	MTBF
Definition	Mean Time Between Failure quantifies asset Reliability and represents the average Utilised Time (UT) between Unscheduled Maintenance (UM) events.
Performance Indicator	Increase in MTBF Annually reported monthly.
Calculation	Utilised Time (UT)/# Unscheduled Maintenance Events (UM).
Example	Utilisation time taken from delay accounting systems and the number of unscheduled maintenance events from SAP.
Rationale	Provides a means by which equipment reliability is measured. An increase in MTBF indicates that failure frequency is reducing, reliability is improving and is directly related to equipment Availability improvement.
Action	Decrease in MTBF to critical (A) assets to trigger an investigation and action plans implemented for improvement.

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Metric	Availability
Definition	Percentage of Calendar Time an asset or plant was physically available to be used.
Performance Indicator	Site Specific Target in line with Life of Asset Plan.
Calculation	Available Time (AT)/Calendar Time (CT).
Rationale	Availability is the measure of time loss and is directly related to the Reliability of an asset or system and is the probability that an asset or system will be available to perform its function when called upon to do so. A high availability is required to ensure that Glencore Copper meets its production targets.
Action	Decrease in Availability to critical (A) assets to trigger an investigation and action plans implemented for improvement.

Metric	Effective Utilisation
Definition	Percentage of Calendar Time an asset or plant area was performing its primary function.
Performance Indicator	95% of time performing primary function when available.
Calculation	Operating time/Calendar time.
Rationale	Derived from Glencore Cu Time Usage Model and Global KPI's.
Action	Decrease in Effective Utilisation to critical (A) assets to trigger an investigation and action plans implemented for improvement.

Metric	% Planned Work Order Actual Cost
Definition	The total actual costs for all planned work order types compared to the total actual costs for all work orders reported weekly.
Performance Indicator	80% Total Actual Planned Costs.
Calculation	Total Actual Planned Work Order Costs/Total Work Order Costs.
Rationale	Planned work is identified by work order priorities 3, 4 and 5, therefore the actual cost of these planned work orders can be reported, these costs can then be reported against the total actual cost of all work orders. The difference between the total actual cost and the total planned actual cost provides a representation of breakdown and break in costs on work orders with priorities 1 and 2.
Action	Conduct deeper review against cost and work centres to understand where the discrepancies have occurred and put actions in place to close the gaps.

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7. Improvement

The process to review and continually improve the approach and deployment for Asset Management Planning is scheduled Annually, or when there is a sudden change in the Operating Context. As the organization matures a longer review period could be considered.

8. References

The following documents provide detailed support material that should be read in conjunction with this Standard to provide a comprehensive understanding of Asset Management Planning.

TABLE 8-1 REFERENCES

DOCUMENT TITLE	DESCRIPTION
Asset Management Framework	This document defines the overall direction of asset management across Glencore Copper.
International Infrastructure Management Manual (IIMM) 2015	This document provides information regarding "How to" develop and implement Asset Management within asset intensive organisations.
Institute of Asset Management SSG 1,2 and 5 Asset Management Policy Strategy and Plans	This document provides guidance on the development of Asset Management Policy, Strategic Asset Management Plans and Asset Management Plans

9. Revision History

VERSION NO.	REASON FOR REVISION	DATE
0.1	New Manual to support the Asset Management Framework	29/04/2021
0.2	Gather Documents and Create Draft Manual	23/08/2021
0.3	Review and Signoff 1st Draft Manual with Sites	01/09/2021
0.4	Reformat and Issue for Final Review	22/11/2021
1.0	Update Site Signoff and Publish	30/05/2022

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10. Appendix A – Stakeholder Signoff

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
Anthony Green	Asset Management - Australia	Anthony.green@glencore.com.au	21/6/2022
Akeem Araoye	Asset Management – North America	Akeem.Araoye@glencore.ca	21/6/2022
Yuri Yurac	Asset Management – South America	Yuri.yurac@glencore.cl	23/6/2022
Lawrence Hlatshwayo	Asset Management – Africa	Lawrence.hlatshwayo@glencore.co.za	21/6/2022
Rohan Reisener	Asset Management - Australia	Rohan.reisener@glencore.com.au	21/6/2022
Dino Moll	Asset Management - Australia	Dino.Moll@glencore.com.au	21/6/2022

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